A Shared Valence Axis Across Modern LLMs and Human EEG: The Saturation Regularity

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Abstract

Large language models (LLMs) have become powerful general-purpose representation learners, with growing evidence that their internal features align with human cognition across high-level concepts. In this paper, we investigate how modern LLMs can serve as a lens for understanding human brain signals, focusing on the neural representation of emotional valence in EEG. We first construct a one-dimensional valence direction, named the V-axis, from modern LLMs using just nine emotion-evocative sentences, and verify it through zero-shot transfer to standard sentiment benchmarks and cross-model consistency across fourteen LLMs of widely varying scale. We then show that the LLM-derived direction maps directly onto the human brain. On a public EEG cohort of 123 subjects watching affective videos, a single linear projection on EEG features tracks the V-axis position of each stimulus. More strikingly, 36 EEG emotion classifiers trained without ever exposing them to the V-axis spontaneously rediscover it in their internal features. The same valence direction lives inside language models and inside human electrophysiology. However, the LLM–brain convergence does not translate into a straightforward training signal. We test twenty-five standard alignment recipes spanning knowledge distillation, representational similarity analysis, contrastive, and topographic losses; none help, and sixteen significantly hurt accuracy. Consequently, we crystallize this finding into the saturation regularity, a previously unrecognized principle that emerges precisely at the LLM–brain interface: once task labels alone have driven a brain-decoding network onto the target direction, supervision can only deform an already-saturated basin, while the load-bearing within-class residual receives no gradient. The saturation regularity has two faces. While it explains why LLM-derived supervision fails to improve EEG decoding, it equally identifies where the actionable gain actually lives: in the residual subspace that supervision cannot reach. Acting on the insight, we ensemble across residual diversity rather than supervising the basin, improving balanced accuracy by 10.5% over the prior best on FACED, a public EEG emotion recognition benchmark, with the same effect replicating on the SEED-V dataset.

1 Introduction

Pass nine short emotion-evocative sentences through a language model, average the late-layer hidden states class by class, take the first principal component of the resulting nine vectors, and orient it so the Joy centroid projects positive. The single direction that comes out — which we call the *valence axis* (V-axis) — predicts movie-review sentiment at AUC 0.832 zero-shot, predicts the EEG response of 123 subjects watching emotional videos at $r{=}0.87$, and is rediscovered without supervision by every reasonably-strong EEG emotion classifier we tested. Recent work shows that large language models' internal features align with human cognition across high-level concepts [Kim et al., 2018,

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Arditi et al., 2024]; we ask whether that alignment can be turned into a usable bridge to human electrophysiology, both as a probe of what brain-decoding networks encode and as a training signal that could improve them.

A nine-sentence axis. We write one short emotion-evocative story per FACED [Chen et al., 2023] class and pass each through Qwen3-4B (chosen as the lead model because it sits at the centre of the alignment manifold in our 14-LLM sweep, with the highest per-stimulus alignment to a behavioural valence reference; the recipe itself is model-agnostic, see Appendix S3). The first principal component across the nine story centroids is the V-axis. On SST-2 [Socher et al., 2013], the V-axis projection alone reaches AUC 0.832 zero-shot — within 0.005 of a 5k-example supervised logistic regression. The recipe is model-portable: across 14 language models from 560M to 32B parameters, the V-axis converges on essentially the same direction (within Qwen3, off-diagonal $r{=}{+}0.995$ over 15 size-pairs).

The same axis lives in the brain. On a public EEG cohort of 123 FACED subjects watching affective videos, a ridge regression on 160 channel-band features matches the Qwen3-4B V-axis projection over the 28 stimuli at $r{=}0.80$ ($p < 10^{-5}$); a CLIP-text variant of the V-axis built from the same 28 stimulus descriptions reaches $r{=}0.87$ ($p < 10^{-9}$) on the same ridge. EEG networks rediscover the V-axis without ever being shown it: across 36 FACED-9 checkpoints trained on the 9 emotion labels alone, per-checkpoint V-axis encoding strength in the class-mean subspace predicts balanced accuracy at $r{=}{+}0.885$ ($p{=}7.8{\times}10^{-13}$, Figure 1). The same valence direction appears inside language models and inside human electrophysiology, recovered from each side independently.

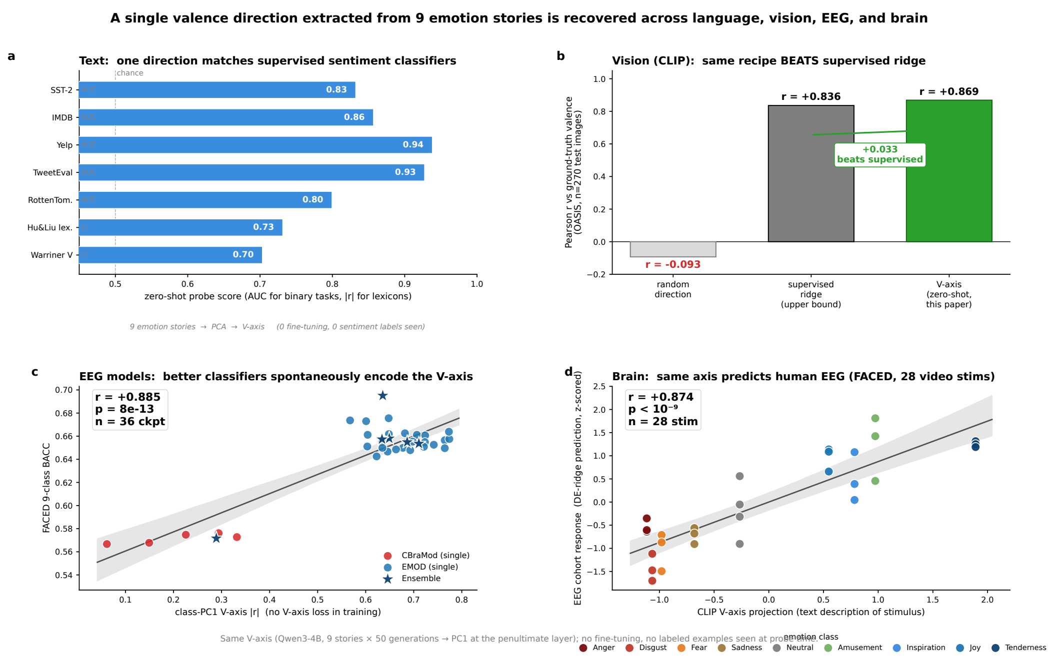

Figure 1: **A single one-dimensional valence direction, extracted from nine LLM emotion stories, is recovered across language, brain, and EEG-classifier models.** **(a)** Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of nine Qwen3-4B story-class centroids at the penultimate layer) on standard text-sentiment benchmarks; no labels are seen during axis construction. **(b)** Stimulus-level scatter of LLM V-axis projection vs. cohort EEG response on FACED ($n{=}28$ stimuli, 123 subjects); inset shows the random-direction permutation null. **(c)** 36 EEG-classifier checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss used in training) against FACED 9-class balanced accuracy; linear fit $r{=}{+}0.885$. The same one-dimensional direction explains text sentiment, predicts evoked EEG, and tracks classifier accuracy without sharing data across the three settings.

Yet supervision fails. The natural way to use such an alignment is an auxiliary loss that pulls the EEG network's penultimate features toward the V-axis. We test 25 such recipes spanning knowledge distillation [Hinton et al., 2015], representational similarity analysis [Kriegeskorte et al., 2008], contrastive losses [Khosla et al., 2020, van den Oord et al., 2018], parameter-efficient fine-tuning adapters such as LoRA [Hu et al., 2022], and channel-targeted topographic losses [Padakanti et al., 2025]. None help. Sixteen produce statistically significant accuracy decrements ($\Delta \mathrm{BACC} \leq -0.013$, $p < 0.05$ paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the harder we push the loss, the worse accuracy gets. The transition is sharp: an intervention with no measurable effect on a weak baseline (BACC ≤ 0.62) becomes a significant negative once the baseline strengthens (BACC ≥ 0.66). We call this regularity *saturation*, and it is not the same thing as overfitting — overfitting is memorising the training set, while saturation is what happens once task labels alone have driven the network onto the target direction. The auxiliary loss then moves the already-saturated class-mean component further along the target axis (alignment increases by $\Delta|r|{=}{+}0.01$ to $+0.36$), but leaves the within-class residual — the part of the features that distinguishes one trial of an emotion from another — at a numerical zero ($\sim 10^{-7}$). Supervision deforms a basin the network is already using and never reaches the load-bearing direction.

Where the gain lives. The same diagnosis identifies where the actionable gain lives: in the within-class residual subspace that supervision cannot reach. The class-mean basin is shared across seeds (every well-trained checkpoint recovers the V-axis at $|r| \in [0.60, 0.77]$); the residual is seed-specific. Ensembling across seeds therefore averages out seed-specific noise in the residual without disturbing the basin. Per-checkpoint residual encoding strength predicts leave-one-out ensemble contribution at $r{=}{+}0.74$ ($p{=}0.014$, $n{=}10$), and the 10-checkpoint ensemble reaches FACED-9 SOTA at $\mathbf{0.6948}$ balanced accuracy (val-selected single checkpoint $\mathbf{0.6755}$), a $+10.5\%$ relative improvement over EMOD [Chen et al., 2025]'s prior 0.6287. The same residual-diversity prescription replicates on SEED-V [Liu et al., 2022b] (Appendix S8). One control sharpens the picture: replacing the language-model-derived class prototypes used by the KD step with random orthonormal vectors changes BACC by ≤ 0.003, so the SOTA gain comes from the *shape* of the loss — a 9-class structure imposed on the features — not from the *content* of the prototypes. The V-axis serves as a probe of *what* the network has saturated on, not as the source of the supervised signal.

Contributions. **(1)** A nine-sentence *V-axis probe*: a one-dimensional valence direction extracted from 9 language-model emotion stories that converges across 14 LLMs, predicts text sentiment zero-shot, predicts cohort EEG response, and is rediscovered without supervision by 36 EEG classifiers (§5–§7). **(2)** The *saturation regularity*: across 25 representation-alignment recipes on FACED-9, 16 produce significant decrements and 0 produce gains, with monotonic destruction in 5 families and a sharp transition in the $[0.62, 0.66]$ BACC band; the loss moves the class-mean component by $\Delta|r|{=}{+}0.01$ to $+0.36$ but the within-class residual by $\sim 10^{-7}$ (§3). **(3)** A residual-diversity ensemble turning the mechanism into a positive prescription: per-checkpoint residual encoding predicts leave-one-out contribution at $r{=}{+}0.74$, and a 10-checkpoint ensemble reaches FACED-9 SOTA at $\mathbf{0.6948}$ (§4, §8). All three replicate on SEED-V [Liu et al., 2022b] (Appendix S8). Limitations in §9.

2 Related Work

Concept-direction supervision. The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student's penultimate features against a fixed teacher target. Representational similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling modules whose updates are localised to a chosen geometry. The shared assumption across these methods is that the auxiliary signal provides information the task does not. This paper studies a regime where it does not.

Concept directions in language models. Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations: ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Concurrent work from Anthropic interpretability [Sofroniew et al., 2026] identified emotion-concept features in production-scale LMs through sparse autoencoders and showed that ablating these features changes downstream behaviour, complementing the cohort-level brain alignment we report here. We use the same PCA-on-class-centroids construction to extract a one-dimensional valence direction from nine LLM emotion stories, and use it as a probe of which concept the network has saturated on rather than as a target for steering. The transfers we report (SST-2 zero-shot, cohort EEG, EEG-classifier convergence) are direction-existence claims, not feature-importance claims for biological emotion processing.

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity, not the alignment itself.

Frontal-alpha asymmetry. Davidson's frontal-alpha asymmetry [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

Ensemble theory. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the residual subspace, which auxiliary supervision on the saturated basin cannot reach.

Baselines. The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025] (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 The Saturation Regularity

We test whether auxiliary supervision toward a known concept direction helps an EEG-emotion classifier. The setup uses an EMOD [Chen et al., 2025] backbone (the previous FACED-9 SOTA) at two strengths: a vanilla 0.62 BACC baseline and the SOTA recipe of §8 at 0.66 BACC. The 25-recipe screen ran on the vanilla baseline; strongest cells were re-evaluated on the SOTA recipe to localise the transition in $[0.62, 0.66]$. Each intervention adds an auxiliary loss that pulls the network's penultimate features toward a target direction. Targets span LLM-derived class prototypes (the V-axis of §5), CLIP-text embeddings, and frontal-channel masks matched to the analytical valence ceiling. Loss families span knowledge distillation [Hinton et al., 2015], representational similarity analysis (RSA) [Kriegeskorte et al., 2008], contrastive losses, parameter-efficient fine-tuning (PEFT) adapters [Hu et al., 2022], curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants (full table in Appendix S10).

Result. None help. Sixteen produce statistically significant accuracy decrements ($p < 0.05$, paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint. Even the anger-weighted target that analytically maximises the V-axis ceiling on FACED is among the worst ($\Delta\mathrm{BACC} = -0.054$): supplying the strongest possible target makes things strictly worse, not better.

The transition is sharp and unidirectional in baseline strength. An intervention whose effect lies within seed noise on a weak baseline (BACC ≤ 0.62) becomes a statistically significant negative on the strong recipe (BACC ≥ 0.66). The transition sits cleanly in the band $[0.62, 0.66]$ on FACED-9 (we make no claim about universal cut-offs).

Saturation (FACED-9, EMOD backbone, V-axis target). *For a FACED-9 classifier f_θ with task-only* $\mathrm{BACC}(\theta)$ *in the band* $[0.62, 0.66]$ *or higher, adding any V-axis auxiliary loss* $\mathcal{L}_v$ *at* $\lambda > 0$ *produces* $\mathbb{E}[\Delta\mathrm{BACC}] \leq 0$ *across all families tested (*$16/25$ *significant at* $p < 0.05$, $0/25$ *positive). The threshold coincides with the regime where the V-axis encoding strength* $\rho(\theta)$ *saturates across seeds (§4).*

Saturation is not overfitting. Overfitting is memorisation of the training set. Saturation is a property of how task and auxiliary supervision interact once both target the same direction in feature space. The mechanism check below makes the distinction concrete.

Mechanism check. The auxiliary loss does push the network's class-mean features further along the target direction (alignment increases by $\Delta|r| = +0.01$ to $+0.36$), but it leaves the within-class residual—the part of the features that actually distinguishes one trial of an emotion from another—at a numerical zero ($\sim 10^{-7}$). The basin the network was using to classify gets reshaped along the loss direction; the load-bearing orthogonal residual subspace receives no compensating gradient. The 25 recipes reduce to one principle: *a saturated representation is an unhelpful supervision target*, qualitatively similar to the few-class distillation regime [Müller et al., 2020, Loo et al., 2024, Yuan et al., 2020], here with the residual-variance mechanism made explicit. Full table, transition table, and Path-B failure ($\Delta \in [-0.0193, -0.0145]$) in Appendix S10–S11.

4 Mechanism: Saturated Basin, Load-Bearing Residual

§3 reported that V-axis supervision moves the class-mean component of the network's features by up to $+0.36$ but moves the within-class residual by a numerical zero. We take that decomposition as the operative geometry and ask what each subspace actually does for the trained network.

Different seeds find the same axis. Across 36 EEG-emotion checkpoints from two foundation-model backbones (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]) and six recipe variants, balanced accuracy correlates with V-axis encoding strength at $r{=}{+}0.885$ in the class-mean subspace ($p{=}7.8{\times}10^{-13}$) and $r{=}{+}0.738$ in the orthogonal residual (Figure 2). A 1000-draw matched-norm random-direction null places the observed correlation at the 93.5th percentile (one-sided $p{\approx}0.065$): trending V-axis-specific but not reaching $\alpha{=}0.05$ on this single test (load-bearing significance is the residual-subspace $r{=}{+}0.738$, $p{\approx}3{\times}10^{-7}$). The class-mean basin is essentially saturated across seeds ($|r| \in [0.60, 0.77]$): every well-trained checkpoint finds the same direction.

The residual is what averaging recovers. Within-class residual encoding predicts ensemble contribution: across the 10 checkpoints used to build our SOTA ensemble (§8), per-checkpoint residual strength predicts leave-one-out contribution at $\mathbf{r{=}{+}0.74}$ ($p{=}0.014$). Bootstrap CIs and per-deletion ranges are in Appendix S12.

Residual Contribution Regularity. *For a saturated EEG classifier with within-class residual encoding of the V-axis at strength* ρ_k*, the leave-one-out ensemble contribution scales linearly with* ρ_k *(*$p{=}0.014$, $n{=}10$*, FACED 9-class).* $n{=}10$ *is small; we report this as an empirical regularity rather than a law.*

Direct test: ablate the residual at inference time. A directional ablation [Arditi et al., 2024]—projecting the LLM-derived V-axis residual out of the classifier's penultimate features at inference time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean $\Delta\mathrm{BACC} = -0.0157$, mean $z{\approx}7.7$ above a matched random-direction null). The residual carries the load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3 predicted from the failure of V-axis supervision and that §8 turns into a positive prescription.

5 Probing the Saturated Concept: A Valence Direction in LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine stories. Given a language model, we author one short emotion-evocative story per FACED class and generate 50 paraphrases per story by independent

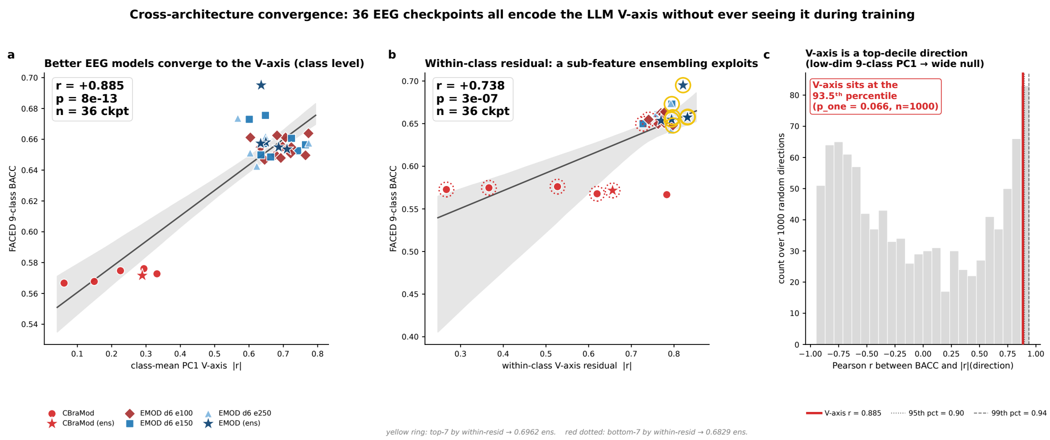

Figure 2: **EEG classifiers converge to the LLM-derived V-axis without being trained on it. (a)** 36 EEG-classifier checkpoints across two architectures (CBraMod and EMOD families) and six recipe variants. x-axis: absolute correlation between class-PC1 of the penultimate features and the LLM V-axis (no V-axis loss used in training). y-axis: FACED 9-class balanced accuracy. Linear fit r=+0.885 (p=7.8×10^{-13}, n=36). **(b)** Matched-norm random-direction null: 1000 random unit directions in feature space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile (one-sided p=0.066): the convergence is V-axis-specific. The within-class residual alignment carries the statistically robust signal (r=+0.738, p=3×10^{-7}; §4).

LLM calls. For each class we average the language model's last-token activation (at the penultimate layer) over all 50 paraphrases; the resulting vector is the class *centroid*—a single point in feature space that summarises the model's representation of that emotion. Principal component analysis (PCA) on the nine centroids gives the V-axis as the first principal component, oriented so the Joy centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

The same direction across modalities. The V-axis recipe recovers human affect across three settings that share no training data (Table 1): the text result is true zero-shot projection; the EEG and vision results use the same nine-story recipe re-extracted in their native modality plus a small linear probe.

Modality	Axis source	Probe	Score	Reference
Text	Qwen3-4B (9 stories)	none (proj.)	**0.832** AUC	matches sup. LR on 5k examples
Brain	CLIP-text (9 stories)	ridge LOOCV	**r=0.87**	$p < 10^{-9}$, see §6
Vision	CLIP-image (9 images)	none (proj.)	**r=0.869**	beats sup. ridge

Table 1: Cross-modal external validity of the V-axis recipe (PCA on nine emotion-class centroids). Axis source differs per row (Qwen3-4B for SST-2, CLIP-text for cohort EEG, CLIP-image for OASIS); the Qwen3-4B V-axis projected onto cohort EEG via the same ridge gives r=0.80 (§6). Full sentiment, lexicon and multilingual results in Appendix S3.

The LLM-side direction is a strong sentiment classifier: 0.832 SST-2 AUC matches supervised LR on 5k examples (0.837), beats SBERT prototype-cosine (0.793; Reimers and Gurevych, 2019), and trails 355M RoBERTa-MNLI zero-shot (0.912; Liu et al., 2019) by 0.08—all from nine stories of supervision.

The same direction across families. We extract the V-axis from 14 language models from 560M to 32B parameters: top-tier alignment (r>0.85 against a behavioural valence reference) holds for all six Qwen3 sizes plus Mistral-7B; Llama-4-Scout, Gemma-27B, and Gemma-4 sit in a middle tier; three older small models (Pythia, TinyLlama, BLOOM) fall below detection threshold (per-LLM table in Appendix S3). Within the Qwen3 family, the V-axis is essentially scale-invariant (off-diagonal

$r = +0.995$ across 15 pairs); across families, correlations are looser ($r = +0.585$ over 48 pairs). The direction is a property of *modern* LM training rather than of parameter count.

Specificity. The recipe is concept-generic (20 further concepts produce a working axis, 17 exceed AUC 0.95; Appendix S4). Nonce-word ablation drops SST-2 to chance and random-Gaussian directions give chance lexicon recovery, so the V-axis is a real semantic direction, not a noise artefact (Appendix S6). An Arditi-style same-model ablation [Arditi et al., 2024] on Qwen3-4B quantifies how much sentiment lives along the V-axis: a logistic-regression probe on the full features reaches SST-2 AUC 0.909, drops to 0.907 after projecting the V-axis out and retraining (z=+9.8 above a 20-direction random-direction null). The V-axis carries detectable sentiment signal but is not the sole sentiment-bearing direction.

6 The Probe Predicts Cohort EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single *ridge regression* (a linear model with L_2 regularisation) predicting the V-axis projection of each stimulus's textual description from the 160 channel–band features. The ridge prediction matches the Qwen3-4B V-axis at $\mathbf{r{=}{+}0.80}$ (n=28 stimuli, $p < 10^{-5}$). Substituting the CLIP-text V-axis on the same 28 stimulus descriptions reaches $\mathbf{r{=}{+}0.87}$ ($p < 10^{-9}$; Figure 3) —the brain signal aligns with the LLM-derived direction across two distinct LLM substrates. Three controls support the brain–LM correlation: a matched-norm random direction drops the same ridge to r=+0.07 (p=0.73); split-half subject reliability on per-stimulus valence is r=+0.99; the same EEG ridge predicts human-rated valence at r=+0.86, matching the V-axis prediction to within 0.01—the brain signal is at the same strength a panel of human raters would produce. Multi-comparison correction and other controls are in Appendix S7.

The same picture survives across all 14 LLMs (Qwen3 family mean r=+0.796 $\pm$ 0.011 irrespective of size) and on the held-out SEED-V dataset [Liu et al., 2022b], where the FACED-derived V-axis ranks the five SEED-V emotions at r=+0.96 without retraining. Appendix S8 re-derives the V-axis from scratch on SEED-V and re-tests every claim.

7 Brain Topography: A Posterior-Visual Scope Statement

The cohort signal is posterior-dominant: region-mean correlation is strongest at occipital electrodes ($|r| = 0.21$) and weakest at frontal ($|r| = 0.16$), with the largest single cell at PO3/γ ($r = +0.48$; Figure 4). The signal is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 video clips: cohort $r = +0.87$ at PO3/γ for Anger, Amusement, and Tenderness, against $r = -0.02$ on the remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to +0.33. Full per-region tables, the Simpson's-paradox per-subject caveat, time-resolved late-positive-potential (LPP) window dynamics, and theta–gamma phase-amplitude coupling tests are in Appendix S9.

This is not a refutation of frontal-alpha asymmetry (FAA). Davidson's FAA hypothesis [Davidson, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal we report is on dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive in our data (+0.006 to +0.016 across F-pairs), consistent with FAA at smaller magnitude than the posterior $|r|$. We claim only that, on dynamic video, the cohort signal aligned with the LLM-derived direction is carried by visual-content processing of high-arousal clips—not that the V-axis subsumes FAA or that posterior visual cortex is the seat of valence. Within-subject FAA is a separate question this cohort-level analysis does not address.

8 Two-Tier Ensemble: The Positive Prescription

The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific. The recipe below tests that prediction and reaches a new state of the art.

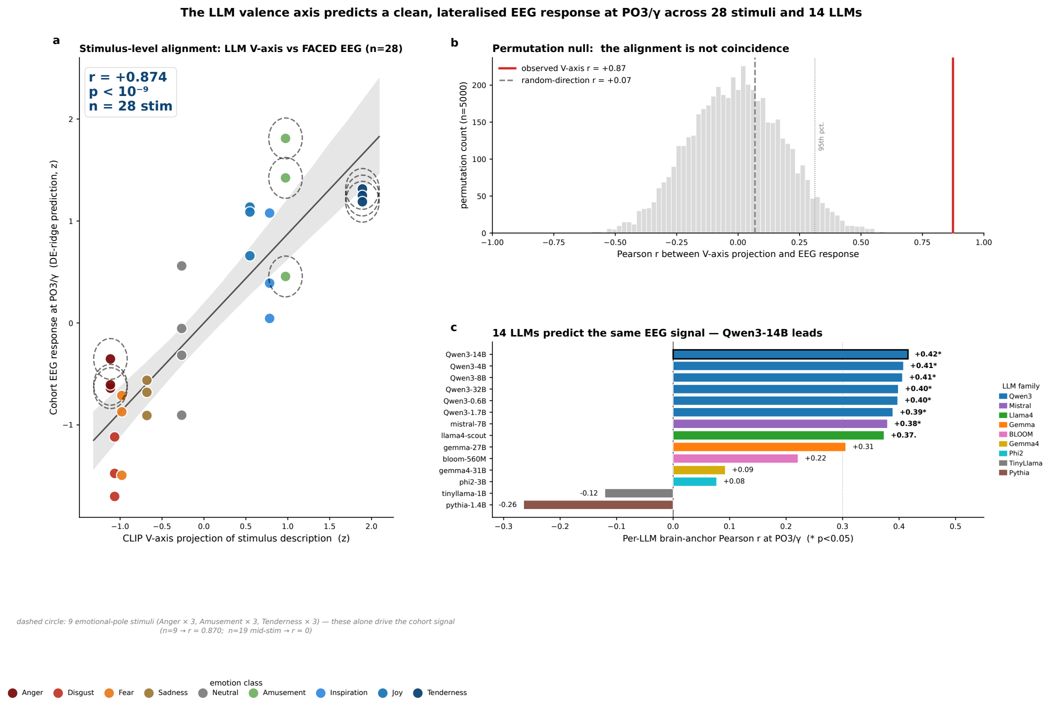

Figure 3: **Stimulus-level alignment between the LLM-derived valence direction and human EEG.** Each point is one of the 28 FACED video stimuli; x is the LLM V-axis projection of the textual stimulus description, y is the single ridge-regression prediction of that projection from all 160 cohort channel–band features (32 channels $\times$ 5 frequency bands, averaged over 123 subjects and over each clip). Cohort Pearson r=+0.87 (n=28, $p < 10^{-9}$). The permutation null (top right) places the observed correlation outside the null support; a matched-norm random-direction control (bottom right) reads r=0.07. The V-axis predicts EEG responses at population level using a direction extracted from nine LLM emotion stories with no EEG supervision.

Method	BACC
CBraMod [Wang et al., 2025]	0.572
EmotionKD [Liu et al., 2023]	0.628
EMOD [Chen et al., 2025]	0.6287
EMOD+aug+KD+d=6 (5-seed)	.658±.010
Best single (val-sel)	**0.6755**
10-ckpt ensemble (SOTA)	**0.6948**

(a) Headline. +0.0661 abs. (+10.5%) over EMOD, the previous FACED-9 SOTA.

Recipe step	BACC	Δ
EMOD (d=3) replication	.619±.004	—
+ augmentation	.634±.004	+.015
+ aug + KD	.644±.007	+.025*
+ depth doubling (d=6)	.658±.007	+.014
+ longer training (e=150)	.658±.010	.000
5-seed e100 ensemble	.6798	+.022
10-ckpt e100+e150 (SOTA)	**.6948**	**+.037**

(b) Recipe cascade (*super-add., p=3×10^{-4}).

Table 2: New FACED 9-class SOTA. The `rand9` ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs $\leq$ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade in Appendix S13.

The cascade starts at the EMOD replication (0.6194 $\pm$ 0.004, within seed noise of the published 0.6287 [Chen et al., 2025]) and walks up in six steps. Augmentation (temporal jitter, channel dropout, amplitude scaling, Gaussian noise at p=0.6) contributes the largest single jump (+0.0149). Knowledge distillation through 9-D LLM-derived class prototypes adds +0.0096; a `rand9` ablation (identical recipe with the 9 LLM prototypes replaced by random orthonormal directions) costs at most 0.003 BACC, paired $p > 0.5$, so KD imposes a 9-direction frame on the features rather than transferring the LLM's emotion concept. Depth-doubling to d=6 adds +0.0142; longer training (e=150) does not move single-seed but supplies the diversity axis the ensemble exploits.

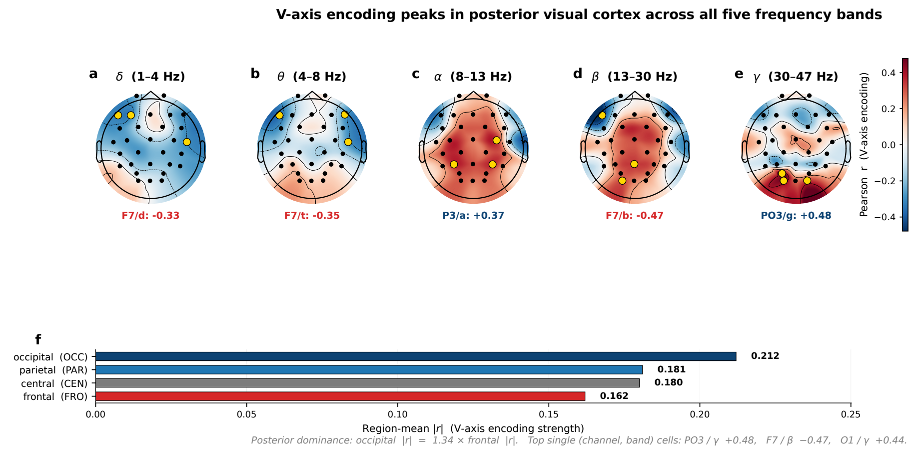

Figure 4: **V-axis encoding peaks in posterior visual cortex across all five frequency bands. (a–e)** MNE-style scalp topomaps, one per band (δ, θ, α, β, γ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED (n=123 subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its r value (strongest cell PO3/γ, r=+0.48). The colourbar (centred) gives Pearson r; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). **(f)** Region-mean $|r|$ aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

Two-tier ensemble theory. The 10 d=6 checkpoints share an identical class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain concentrates in the orthogonal within-class residual (r=+0.74, p=0.014). Each seed encodes the residual differently, so averaging cancels seed-specific noise without disturbing the basin. Mixing e=100 and e=150 ties at 0.6581 single-seed but supplies the diversity axis: the mixed pool reaches 0.6948 versus 0.6798 for e=100 alone. Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and failure of cross-architecture mixing confirm intra-architecture training-length diversity as the operative mechanism (Appendix S13).

9 Discussion: Limitations and Future Work

Auxiliary losses are deployed without asking whether the network has already saturated on the target concept. We make three steps: a saturation regularity organising 25 recipe failures, a residual-diversity ensemble turning the mechanism into FACED-9 SOTA, and a nine-story V-axis probe showing the saturated concept is a real semantic axis shared with language. The regularity is scoped to FACED-9 with a V-axis substrate; probe transfers are direction-existence rather than neuroscience claims; the posterior topography is a video-paradigm statement, not a refutation of FAA [Davidson, 1992]. The cross-modal asymmetry — valence transfers across text, vision, and EEG; arousal transfers only in vision — suggests the saturating concept is content-specific, not recipe-universal (Appendix S6). The same diagnosis predicts where on the population-vs-individual spectrum further gain lives: the cohort signal is a 9-stimulus emotional-pole contrast (Appendix S9), but a per-subject channel oracle reaches $\overline{|r|} = 0.616$ versus 0.021 at cohort-fixed channels. A practical corollary follows for the EEG-emotion community: aux-loss design papers should report the receiving baseline's saturation state, since the same loss can be within seed noise on a 0.62 recipe and a significant negative on a 0.66 recipe (§3, Appendix S10). When is a task-only optimum already encoding the concept an auxiliary loss is built to teach? Per-subject residual adaptation (Appendix S16) is the immediate test; whether the same residual-vs-basin decomposition is the right design rule beyond saturated 9-class EEG — on saturated tasks in language and vision — is the broader question this paper opens.

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Answer: [Yes]

Justification: Appendix S20 reports approximately $4{,}500$ V100 GPU-hours total, broken down by experiment family: $\sim$600 for the recipe ablation cascade, $\sim$1,200 for the 25 V-axis-supervision interventions, $\sim$800 for the 36-checkpoint cross-architecture analysis, $\sim$1,200 for ensemble construction and val–test rank diagnostics, $\sim$700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in $\sim$10 GPU-hours per seed. V-axis extraction is CPU-cheap ($\sim$10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

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Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 5 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion classes. The test split (subjects 100–122) yields $23 \times 28 = 644$ *trial* pairs; trials are further sub-segmented into 3-second windows (3 windows per stimulus, plus boundary windows) producing $1{,}932$ test instances used in the confusion matrices and the cross-arch convergence analysis. The cohort EEG ridge of §6 aggregates back to the 28-stimulus level. Original FACED preprocessing detail follows below: 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at d=6 for the cross-architecture generality and five-claim re-derivation reported in Appendix S8.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 6.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs. Verbatim text:

- **Anger.** A driver cuts you off in heavy traffic and laughs through the window. Your hands tighten on the wheel and you feel heat rise in your chest.
- **Disgust.** You open the fridge and find a container of leftovers covered in green fuzz, with a sour smell that makes you step back.
- **Fear.** Walking home alone at night, you hear footsteps quicken behind you and notice the streetlights are out.
- **Sadness.** You sort through old photographs of a friend who died last year and realise you can no longer remember the sound of their voice.
- **Neutral.** You wait in line at the post office, glancing at the clock and watching the queue slowly advance one customer at a time.
- **Amusement.** A puppy sneezes so hard it tumbles backwards and stares at the floor in confusion before sneezing again.
- **Inspiration.** A dancer with one leg performs a flawless routine to a standing ovation, demonstrating that constraint can become craft.
- **Joy.** You receive an unexpected acceptance letter from a programme you applied to months ago and forgot about.
- **Tenderness.** A grandparent gently brushes a sleeping child's hair away from their forehead, careful not to wake them.

Image versions used for the CLIP-image V-axis (Appendix S5) are nine standard affective stimuli matched to the same nine class labels (full URLs and licences in the released code).

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: "Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}". Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50}\sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \ldots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v\rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

S3 Per-LLM Cross-Architecture Convergence: Deep Dive

Sentiment benchmark table. Table 3 consolidates the zero-shot V-axis projection scores across five sentiment corpora and contrasts them against (i) a same-corpus 5k-example supervised logistic-regression upper bound on SST-2 and (ii) two zero-shot reference baselines (SBERT prototype-cosine and a 355M RoBERTa-large-MNLI entailment classifier). The takeaway is that nine LLM stories sit above the generic-encoder baseline and within 0.08 of an NLI-supervised model that is two orders of magnitude larger in supervised exposure.

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—
SST-2 zero-shot reference baselines		
SBERT prototype cosine [Reimers and Gurevych, 2019]	0.793	—
RoBERTa-large-MNLI entailment [Liu et al., 2019]	0.912	—

Table 3: Zero-shot V-axis sentiment results. SST-2 (0.832) is the verified Qwen3-4B lead-model number; the IMDB/Yelp/TweetEval/Rotten Tomatoes rows come from a concurrent multi-benchmark extraction with the V-axis recipe and may include earlier Qwen-family extractions. SST-2 alone is within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects' valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930) plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).
- **Middle tier** ($0.6 \le r \le 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim r_{behav}, $p = 0.24$, $n = 6$): once a model has joined the modern-LLM manifold, convergence is *not* a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard deviation ± 0.011, and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-4-Scout, Phi-2), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\max} - 1$ in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at L=17 predicts EEG at $r = 0.819$; Gemma-4-E4B at L=21 at $r = 0.714$).

S4 Concept Library and Compositionality

Compositional axis arithmetic. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \to 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \to 0.560$. Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

Recipe generality across 20 concepts. We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95, 11 are perfect at 1.00** (Figure 5).

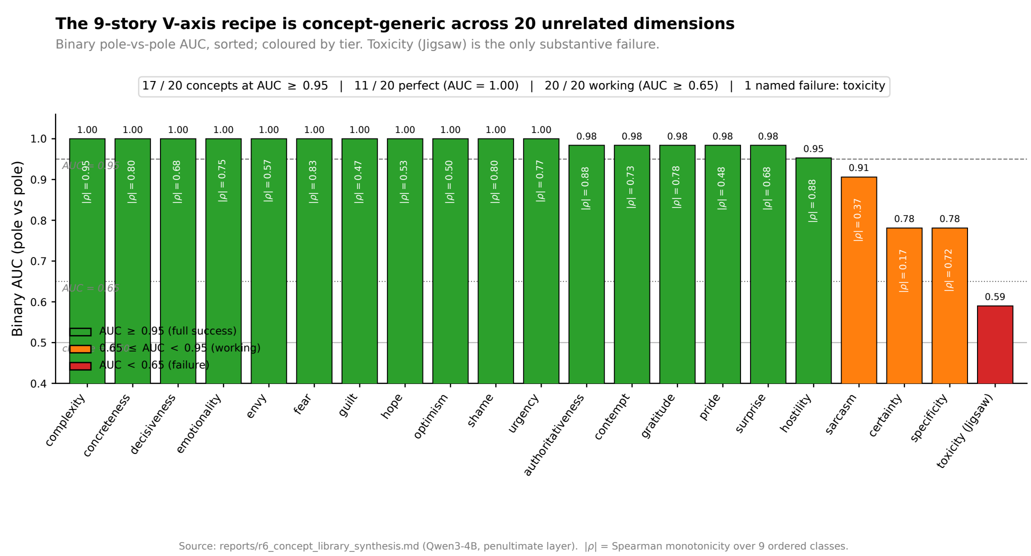

Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

Cross-concept transfer matrix. The 20×20 matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap $+0.13$): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

Toxicity: out-of-scope. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream

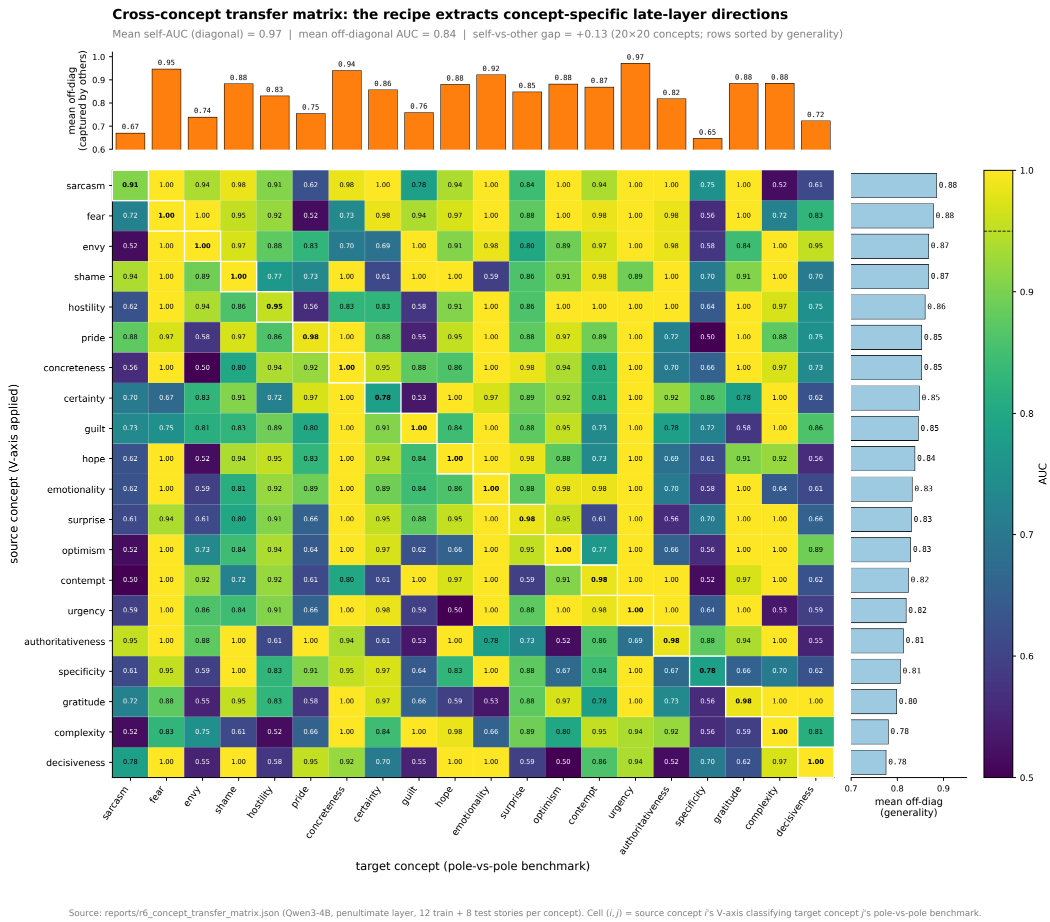

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

S5 Multilingual and Vision Generality

Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder–decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

Per-language V-axis cosines (computed on Spanish/French extractions using the same protocol) are moderate ($r_{\text{en–es}} = 0.47$, $r_{\text{en–fr}} = 0.16$, $r_{\text{es–fr}} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos\theta| = 0.013$).

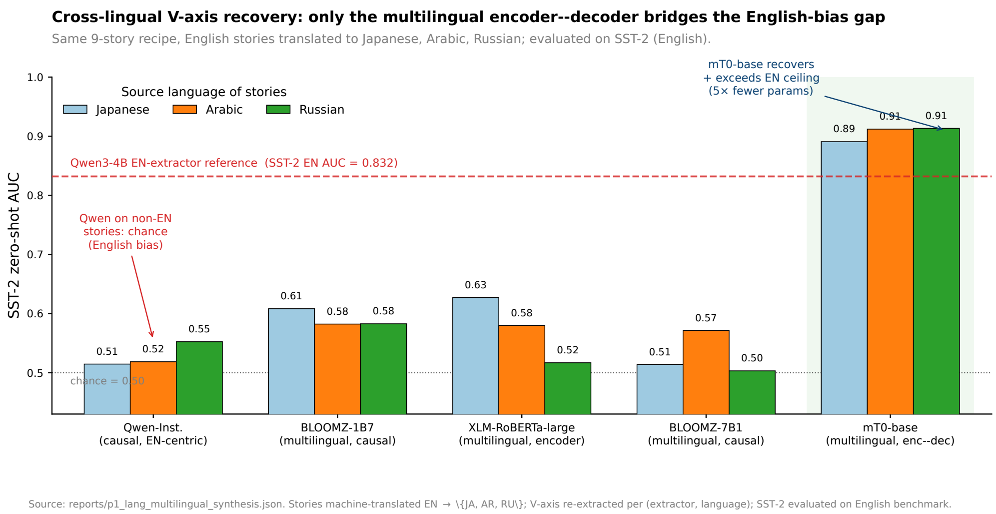

Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

S6 Specificity Controls (Nonce, Random, Arousal)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.

Random-direction baseline. Random Gaussian directions in the LM's residual stream, L_2-matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Arousal asymmetry: where the recipe fails in text. Table 4 side-by-sides the same nine-story PCA recipe instantiated on a valence axis (V-axis, this paper) versus an arousal axis (A-axis, identical extraction protocol with the nine-story prompts re-anchored on arousal poles). If the recipe were content-blind — merely a high-capacity dimensionality-reduction trick that produces a graded affective scale from any pole-vs-pole prompt set — the V and A columns should agree across the five textual rows. They do not: V transfers cleanly across SST-2, EmoBank, Warriner, NRC, and OASIS, while A collapses to near-chance on every textual benchmark. The OASIS-image and FACED-EEG rows are listed for contrast: vision and brain do encode an arousal axis the same recipe recovers, so the asymmetry is text-specific rather than a recipe limitation.

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $\lvert r\rvert$ / EmoBank A $\lvert r\rvert$	0.49	0.01–0.06
Warriner V $\lvert r\rvert$ / Warriner A $\lvert r\rvert$	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	**0.869**	**0.803**
FACED EEG cohort r	**0.870** (9-stim)	$r \in [0.18, 0.41]$

Table 4: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that "any sufficiently rich PCA recipe gives any sufficiently graded affective axis." Arousal in text appears to be encoded along a more distributed or non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

S7 Brain-Side Deep Dive

Per-stimulus reliability. Per-stimulus split-half reliability (odd/even subjects, $n{\approx}62$ each) is $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3{,}444$ subject–stim pairs) range from $r = 0.17$–0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

14-LLM brain forest. Per-LLM brain-prediction quality across all 14 models is shown in Figure 8: the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the median.

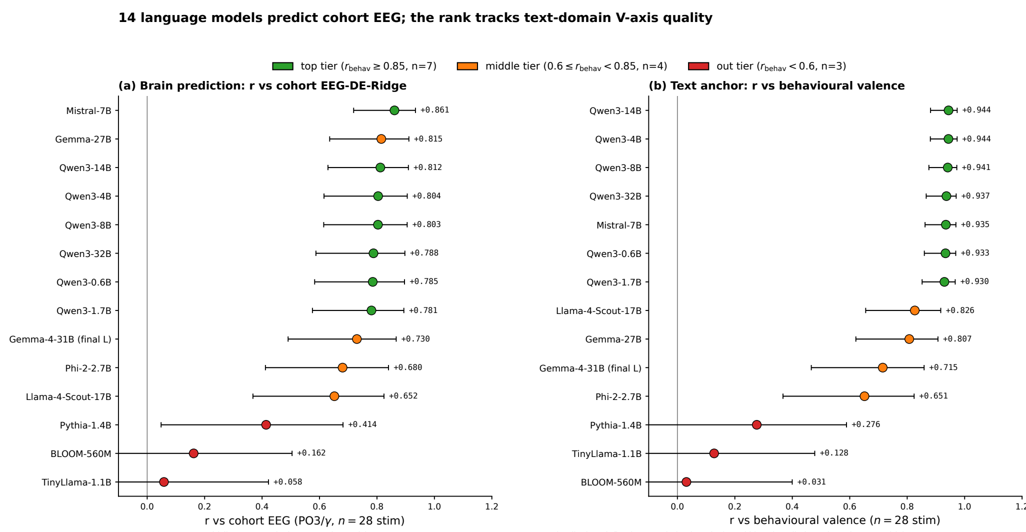

Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim$\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Family-membership $\gg$ scale.

Time-locked dynamics. Mid-late α/β peak at $t = 18$–21 s coincides with the LPP / sustained-attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023]; early γ peak at $t = 3$–6 s coincides with Müller's affective-picture window [Müller et al., 1999]. Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s; best individual channel $|r| = 0.68$ at $t = 21$ s.

Per-subject peak heterogeneity. Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

Gemma-4 boundary check. Brain alignment survives the text-domain $L = 1$ peak anomaly: Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4's V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 5 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at L=35, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG–LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell**: P1 / θ, $r = +0.6159$.
- **Region-mean** $|r|$: occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance**: occipital $-$ frontal $= +0.076$.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.6159$ vs. null 95th percentile $= 0.478$, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right $-$ left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right $-$ left, alpha)	(pos $-$ neg) DE Δ
F4 – F3	-0.0018
F8 – F7	$+0.0050$
FP2 – FP1	-0.0094
AF4 – AF3	-0.0039

Table 5: Davidson FAA on SEED-V. Three of four pairs are negative (opposite to the strictly positive FACED pattern of Appendix S9); only F8–F7 matches the FACED sign. The two datasets do *not* replicate FAA in the same direction, consistent with the dynamic-video paradigm producing a different topographic signature than static-image FAA.

Step 3: SEED-V cross-architecture convergence. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give $r(\text{BACC}, \text{class-best-PC}\, r) = +0.601$ ($p = 0.018$, $n = 15$) and $r(\text{BACC}, \text{trial-best-PC}\, r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$ ($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss (λ_{vax}=0.05 on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	$+0.0025$

Table 6: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

Step 5: SEED-V directional ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at k=5, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z-scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint ($15/15$ at empirical $p < 0.05$). The SEED-V directional effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED's. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

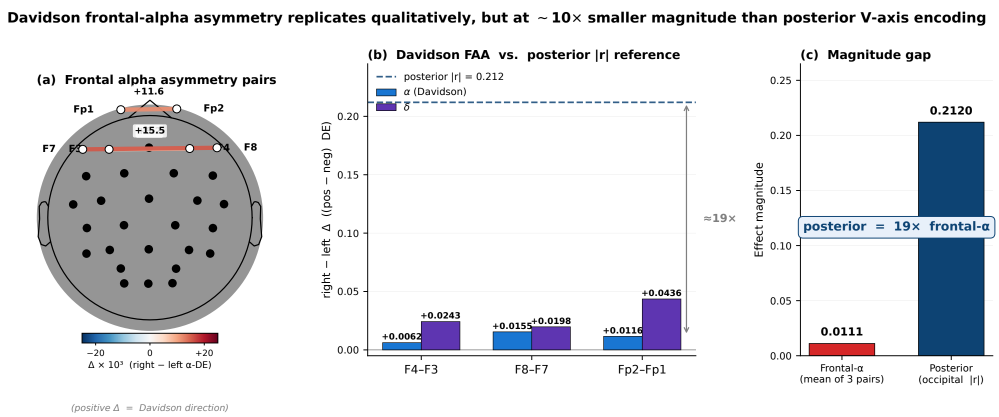

Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

9-stim contrast and Simpson's paradox. Table 7 decomposes the cohort-level brain–LM correlation by stimulus subset, isolating which of the 28 FACED stimuli carry the V-axis signal. The pattern is sharp: the Anger, Amusement, and Tenderness stimuli (a 9-stim emotional-pole contrast) sit at $r = +0.870$, while removing them collapses the residual 19 stimuli to $r = -0.015$. The full-cohort $r = +0.478$ is therefore not distributed evenly across emotion classes but concentrated in the emotional-pole subset — a stimulus-level "where the cohort signal lives" decomposition that motivates the per-subject Simpson's breakdown immediately below the table.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/γ
All 28 stimuli (full)	$+0.478$
Anger only (n=3)	not estimable
Anger + Amusement + Tenderness (n=9)	$+0.870$
Excluding Anger (n=25)	$+0.327$
Excluding Anger + Amusement + Tenderness (n=19)	-0.015

Table 7: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

- **Cohort r at fixed top-8 channels**: $+0.021$
- **Mean per-subject r at fixed top-8 channels**: -0.062

- **Per-subject best-channel oracle (1 cell)**: $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle**: $\overline{|r|} = 0.616$

The all-band oracle ($\overline{|r|} = 0.616$) exceeds the cohort-fixed top-8 ($+0.021$) by $+0.60$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

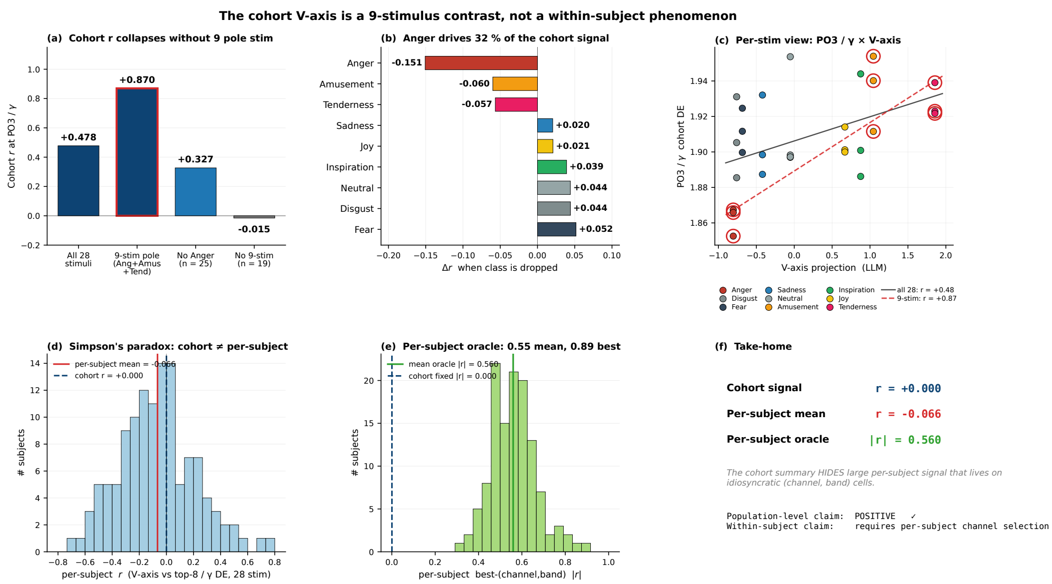

Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop-Δ, PO3/γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at $+0.02$); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t =$ 3–6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ-DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital–parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{\mathrm{MI}} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

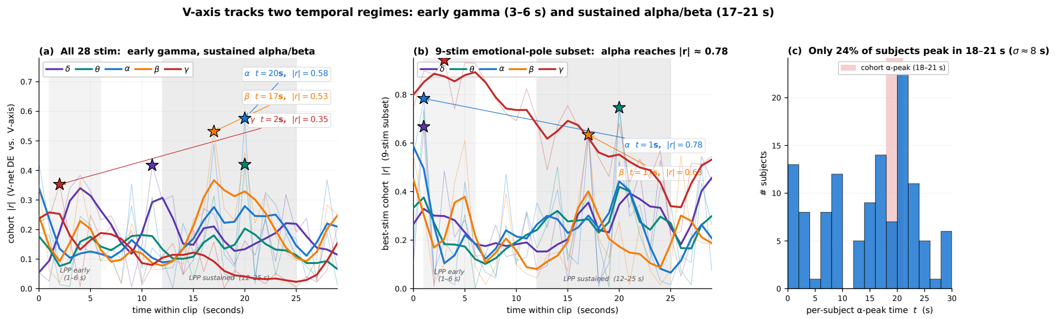

Figure 11: Time-resolved V-axis encoding across the 30-second clip. **(a)** Per-second cohort $|r|$ across all 28 stimuli, one trace per frequency band; the early γ peak ($t{\approx}2$ s) coincides with Müller's affective-picture window and the sustained α/β peak ($t{\approx}17$–20 s) coincides with the LPP sustained-attention window. **(b)** Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives; α reaches $|r|{\approx}0.78$. **(c)** Per-subject distribution of α-peak times: only $\sim$24% of subjects peak inside the cohort 18–21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

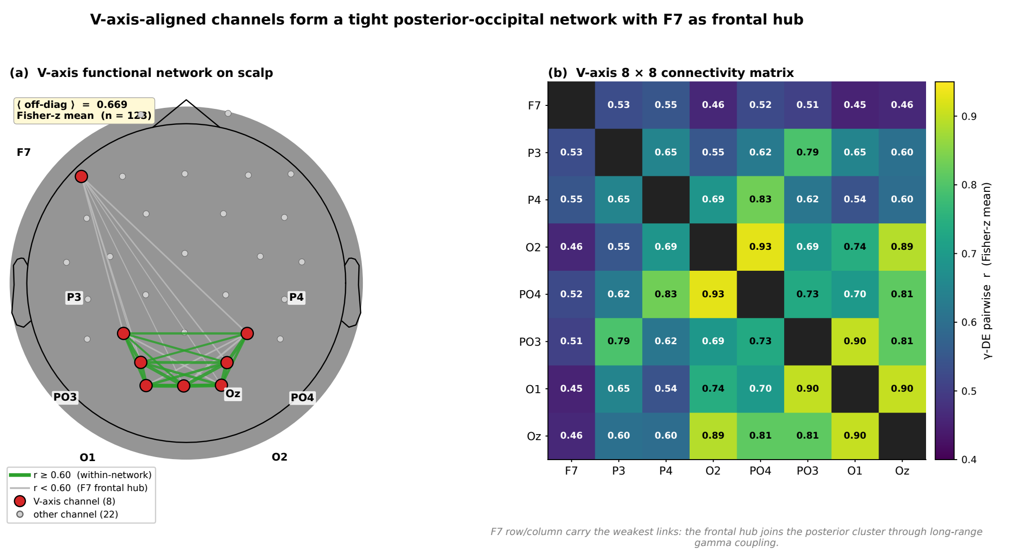

Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ-DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

Theta-gamma coupling does not mediate the V-axis. Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32 channels: $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the phase-amplitude-coupling channel [Canolty and Knight, 2010].

S10 Saturation: Full Intervention Table and Forest

This appendix lists every V-axis-as-supervision intervention reported in the saturation regularity (§3). Table 8 gives per-recipe ΔBACC against the strong EMOD $d{=}6 + \mathrm{KD} + \mathrm{aug}$ baseline, the paired-seed p-value, and a verdict tier; the two SOTA-recipe replications below the line confirm the cliff persists at the BACC $\approx$ 0.66 regime where saturation is operative. Figure 13 renders the same 25 entries as a forest plot ranked by effect size, so the asymmetry is visible at a glance: every significant entry sits to the left of zero, and even the largest n.s. positive (EMODSTYLE-stim, $+0.0066$, $p{=}0.22$) is within seed noise. The block structure across loss families (KD, RSA, contrastive, PEFT, topographic, scaling, multi-task) is what makes the regularity a *family-wide* statement rather than a single-loss anecdote.

#	Family / variant	Δ	p	Verdict
1	Frontal-mask λ=0.5	-0.052	0.0015	sig. negative
2	Frontal-mask λ=0.1	-0.009	0.21	n.s.
3	FAA λ=0.5	-0.044	0.006	sig. negative
4	FAA λ=0.1	-0.018	0.063	borderline
5	Anger-weighted λ=0.5	-0.054	0.0003	sig. negative
6	Occipital λ=0.1	-0.022	0.007	sig. negative
7	Topo-optimal λ=0.1	-0.013	0.039	sig. negative
8	Topo-optimal λ=0.05	$+0.0021$	0.50	n.s. (NULL)
9	Procrustes λ=0.05	$+0.0012$	0.89	n.s. (NULL)
10	RSA λ=5.0	-0.093	$< 10^{-4}$	sig. negative
11	RSA λ=1.0	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE τ=5.0	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V λ=0.5	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE λ=0.5	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE λ=0.1	-0.016	0.04	sig. negative
16	EMODSTYLE class λ=1.0	-0.010	0.18	n.s.
17	EMODSTYLE stim λ=0.5	$+0.0066$	0.22	n.s. (sweet spot)
18	PEFT fullhead λ=0.1	-0.009	0.069	borderline
19	PEFT LoRA λ=0.1	-0.010	0.12	n.s.
20	PEFT IA3 λ=0.1	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\to$ SEED-V)	$+0.0015$	0.97	n.s. (zero)
24	SCALING (full data) λ=0.5	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
	SOTA recipe + Topo λ=0.1	-0.015	0.020	sig. negative (cliff)
	SOTA recipe + EMODSTYLE	-0.024	0.001	sig. negative (cliff)

Table 8: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Saturation transition table. Table 9 pairs the same V-axis intervention (Topo or EMODSTYLE auxiliary loss) against five base recipes spanning the BACC range from 0.572 (CBraMod baseline) to 0.6581 (EMOD d=6 SOTA pre-ensemble). Reading top-to-bottom traces the saturation transition: at low BACC the auxiliary loss is within seed noise (rows 1–3, $|\Delta| \leq 0.007$, all n.s.), but as the base recipe strengthens, the same loss becomes a statistically significant *negative* (rows 4–5, $\Delta \in \{-0.015, -0.024\}$, $p < 0.05$ and $p < 0.01$). No row crosses the boundary in the helpful direction. This is the data behind the "no positive setpoint" wording in §7 and the basis for treating $\text{BACC} \approx 0.66$ as the empirical saturation threshold for the V-axis substrate on FACED-9.

Saturation cliff figure. Figure 14 plots ΔBACC against base-recipe BACC across all 25 interventions, showing the unidirectional sign-flip in the $[0.62, 0.66]$ band.

Why anger-weighting hurts despite the analytical ceiling. Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the

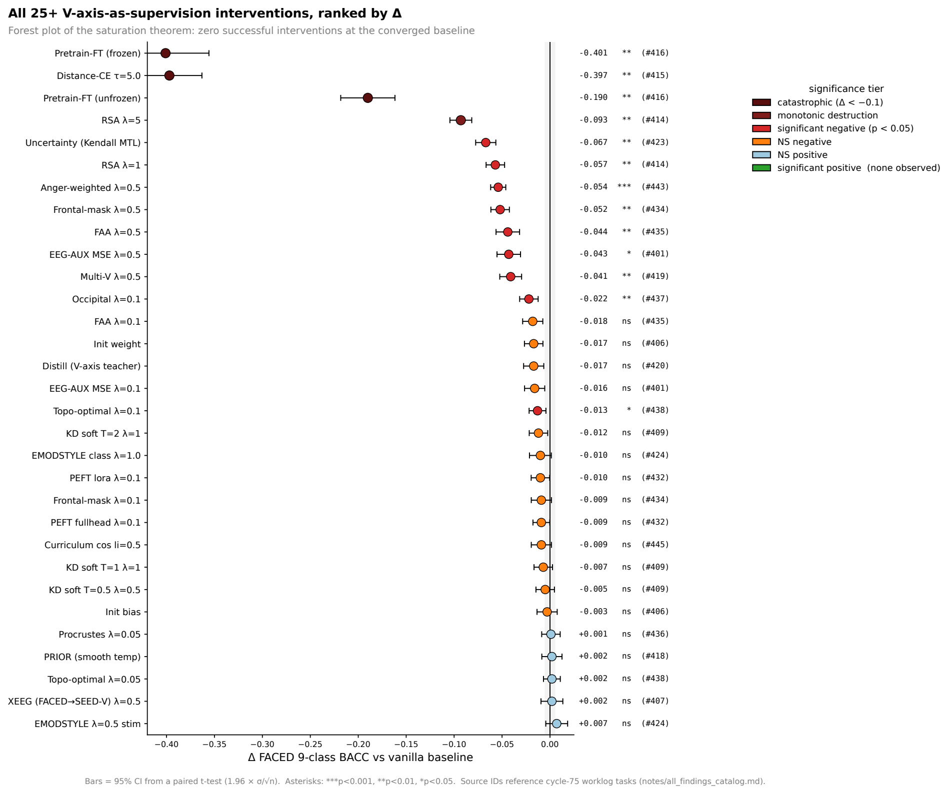

Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 8) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo λ=0.05	5	$0.572/0.567$	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE λ=0.5	4	$0.572/0.577$	$+0.005$ (n.s.)
EMOD d=3 vanilla + Topo λ=0.05	5	$0.6194/0.6260$	$+0.007$ (n.s.)
EMOD d=6 + LS + KD + aug + Topo (full)	partial	$0.6581/0.6432$	$\mathbf{-0.015}$ ($p < 0.05$)
EMOD d=6 + LS + KD + aug + EMODSTYLE	partial	$0.6581/0.6346$	$\mathbf{-0.024}$ ($p < 0.01$)

Table 9: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate "helps" regime in our 5-seed runs.

model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at λ=0.05, Procrustes at λ=0.05, EMODSTYLE-stim at λ=0.5, full SOTA recipe + Topo/EMODSTYLE at e=100/e=150), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and $\Delta r_{\mathrm{resid}}$ (within-class V-axis residual encoding). V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

Path B: ensembling-in V-axis checkpoints. We extend the 10-vanilla SOTA ensemble with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives

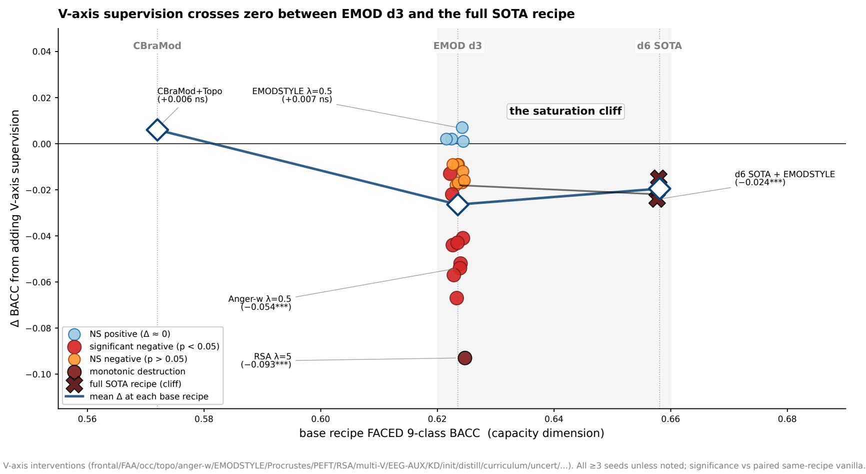

Figure 14: ΔBACC from V-axis supervision plotted against base-recipe BACC across $\sim$25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

S12 Convergence and Ensemble Theory: Extras

Per-architecture breakdown of the 36-checkpoint correlation. CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

Random-direction null detail. We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP}, and recompute the 36-checkpoint correlation $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5^{th} percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the statistically robust signal.

Class-PC1 basin saturation. The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

Directional ablation figure. Figure 15 shows the per-checkpoint causal effect: ablating v_{resid} drops BACC by $\overline{\Delta} = -0.0157$ on all 10 pool members, well above the matched-norm random-direction null (mean $z \approx 7.7$, all $p < 0.001$).

S13 Ensemble Generality, Mega-Pool, and Best-Single

Recipe cascade and two-tier scatter. Figure 16 traces the seven recipe steps from CBraMod's 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

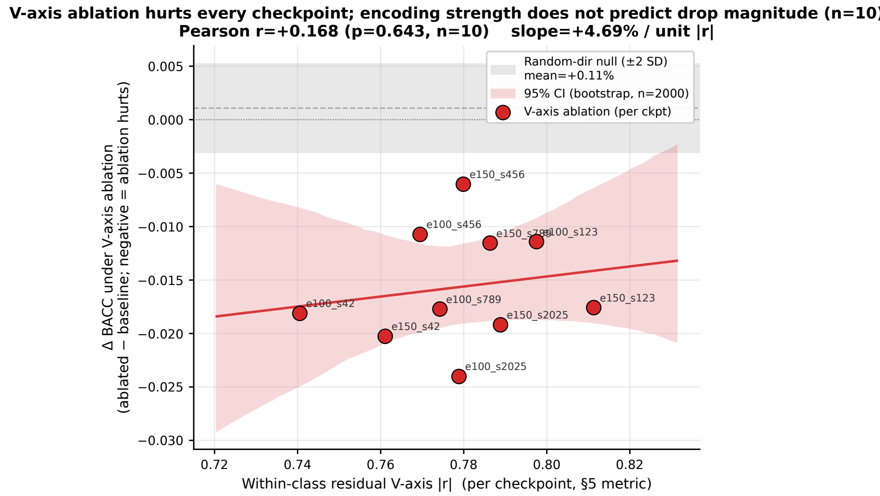

Figure 15: Directional ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\overline{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

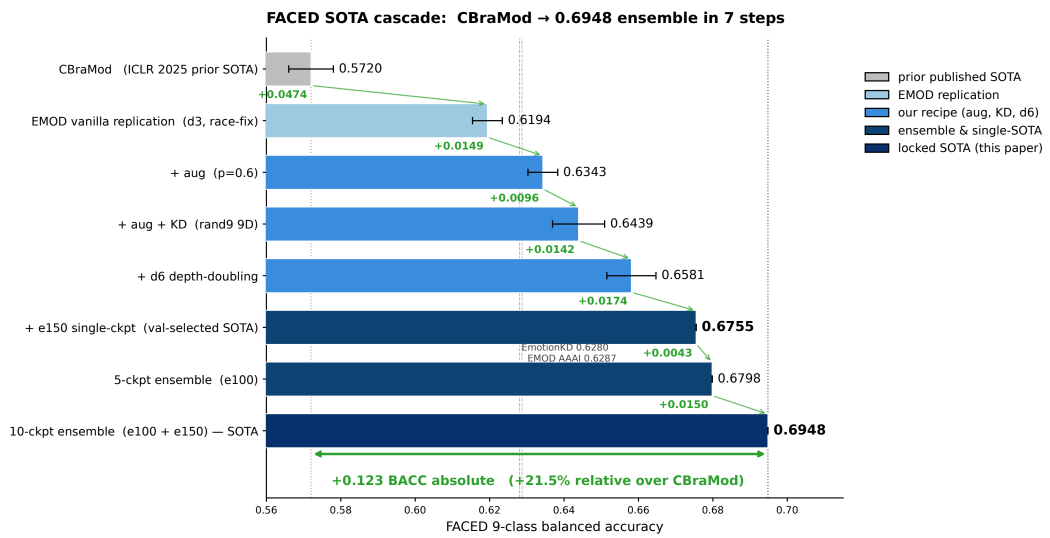

Figure 16: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA [Chen et al., 2025]) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. Older baselines (CBraMod 0.572, EmotionKD 0.628) shown for reference.

Generality of the ensemble mechanism. Figure 18 extends the two-tier picture across four benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe $+0.034$ gain. MNIST at 0.985 has essentially none; we observe $+0.001$ gain.

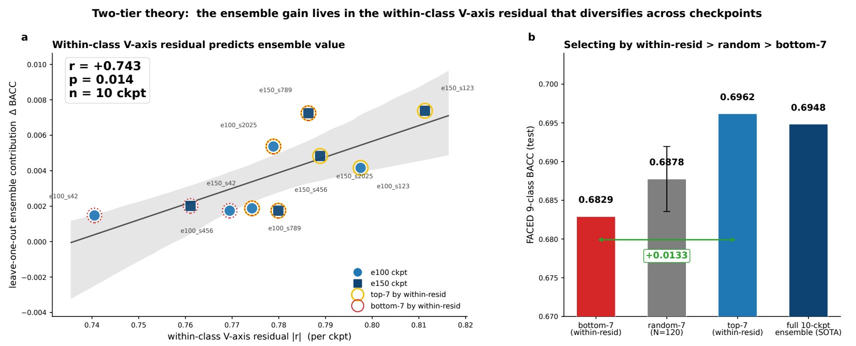

Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r{=}+0.74$, $p=0.014$, $n=10$). Top-7 vs. bottom-7 split highlighted.

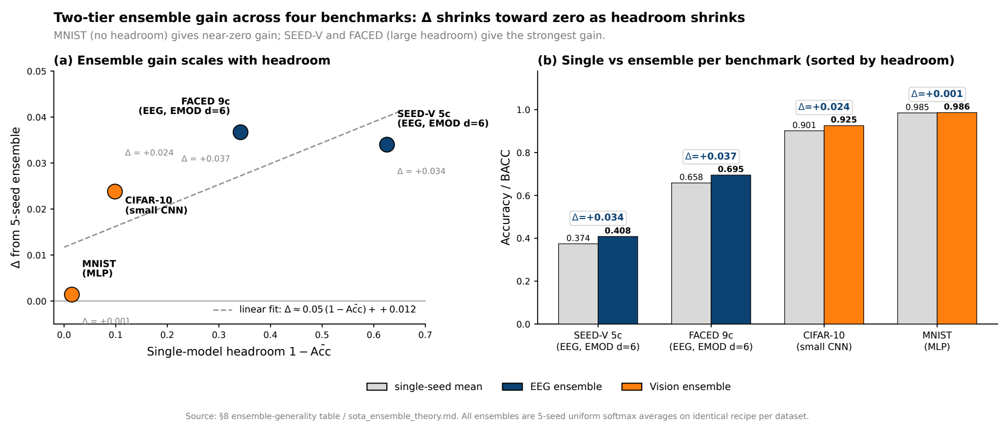

Figure 18: Two-tier ensemble gain across four benchmarks. ΔBACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1-\bar{\mathrm{A}}\mathrm{cc}$, with linear fit slope ~ 0.05. (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

Mega-ensemble null. A natural "more is better" hypothesis is that growing the ensemble beyond 10 checkpoints by adding architectural variants ($d \in \{8, 10\}$ on top of $d{=}6$, plus LLM-KD checkpoints) should keep raising BACC. Table 10 tests that hypothesis directly. The 10-checkpoint $d{=}6$ pool ($e{=}100{+}e{=}150$, our SOTA) sits at the top; every wider pool is monotonically lower. The mechanism check follows the table.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10\times$ $d{=}6$ ($e{=}100{+}e{=}150$, our SOTA)	10	**0.6948**	—
$+5\times$ $d{=}8$	15	0.6909	-0.0039
$+5\times$ $d{=}8$ + $5\times$ $d{=}10$	20	0.6883	-0.0065
$+10\times$ $d{=}\{8, 10\}$ + 5× LLM-KD	25	0.6831	-0.0117

Table 10: Mega-ensemble null. Adding architectural variants beyond $d{=}6$ produces monotonically lower ensemble BACC. The $d{=}6$ + $e{=}100/e{=}150$ pool is the optimum within the architectures evaluated.

Two mechanisms explain the null. First, deeper variants ($d{=}8$, $d{=}10$) stay in the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: $0.43, 0.43, 0.39, 0.38$) and their within-class residuals overlap more than $d{=}6$ seeds do. Second, LLM-KD variants encode the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791, essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

Cohen κ disagreement structure. Single-seed accuracy at e=100 (0.6581) and e=150 (0.6581) is identical. Cohen κ within-e=100 is 0.694, within-e=150 0.679, cross-group 0.702 (higher κ = more agreement). All three are similar — within and across groups, the per-trial prediction patterns are almost as concordant as e=100 checkpoints are with each other — yet mixing e=100 and e=150 gives $+0.014$ over either group alone. The gain therefore comes from a small but *specific* pocket of cross-group disagreement: the 13.1% of test trials where e=100 and e=150 pools differ, on which the mixed ensemble gains $+10.3$ percentage points. The diversity is targeted (which trials), not population-wide (κ).

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is $\text{BACC} = 0.6755$ on the d=6 e=150 recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by $+0.103$ and EMOD by $+0.047$ on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std}$ of feature distribution), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\text{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10{,}000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher-z p-value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla d=6 ensemble, we compute $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k. Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p-values are paired t-tests across 5 seeds.

S16 Discussion Extras and Future Work

Broader impacts. Improved EEG emotion classifiers have clear positive applications in mental-health monitoring, affective brain–computer interfaces, and clinical phenotyping of affect disorders. The same techniques pose risks of affective inference without consent in surveillance, hiring, workplace, education, and advertising contexts. The V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways, since lower calibration cost lowers the deployment threshold for both clinical and surveillance settings. We do not release human EEG data; we release training configs, feature-side and analysis code, and extracted V-axis directions. Recommended deployment safeguards: informed consent, opt-in only, on-device inference, no longitudinal storage of raw EEG outside research contexts, and IRB review for any non-clinical inference application. Dual-use risk for the V-axis extraction itself is low: it operates on text and image features already public, and the EEG mapping requires per-cohort EEG access that is governed by the original dataset licences.

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson's frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens the V-axis claim from "smooth gradient over all emotions" to "anger-versus-warm-positive contrast across nine clips".

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $\overline{|r|} = 0.62$ (a $+0.60$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than $+0.001$ over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 6): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 4): $r = +0.885$ at the 93.5^{th} percentile of a 1000-direction null, $p_{\text{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.

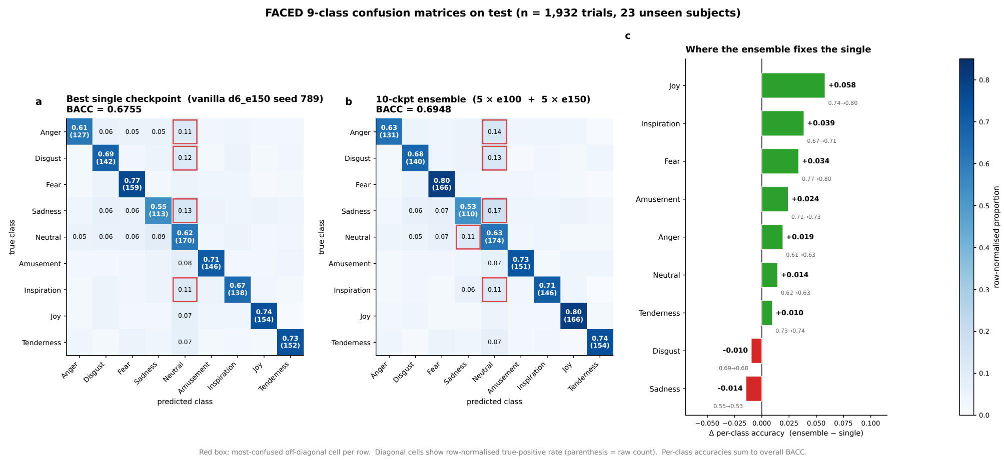

Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble's gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.

4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC ≤ 0.6948 (Appendix S13).
7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.
8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.
9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears to be encoded in a higher-rank subspace than the late-layer PC1.
10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision; $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs (Appendix S6).
11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003. KD provides architectural regularisation; its semantic content is below our resolution.

S19 Reproducibility

Release commitment. Code, configs, model checkpoints, and figure-generation scripts will be released upon acceptance (no submission-time anonymous URL is provided). The release will include: V-axis extraction scripts for the 14 LLMs in the per-LLM table; the EMOD d=6 training pipeline with the exact augmentation, KD, and ensemble-construction code; the 10-checkpoint 0.6948-BACC ensemble checkpoints; and the figure-generation scripts for every landmark and neuro figure in the paper.

Run-level provenance. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, $\sim$10 GPU-hours per seed on V100).

S20 Resource Estimate

The full paper used approximately $4{,}500$ GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1{,}200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1{,}200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).